EE 102 Week 1, Lecture 1 (Fall 2026)

Date: August 31, 2026

Learning Outcomes

By the end of this lecture, you should be able to:

- Understand the meaning of signals and systems
- Connect how the mathematical prerequisites (vectors, complex numbers, matrices) are relevant in EE 102.

Contents

1 Introduction to Signals and Systems

A *signal* is a set of data or information. This is intentionally defined in a very broad manner. A simple way to understand signals: all mathematical functions that you have studied in your calculus classes are signals if you can attach a physical meaning to the function. Note that a signal need not be a function of time. It is often intuitive to think about functions of time and physical signals are functions of time (often, but not always!).

A *system* maps (that is, it processes) input signals into output signals. So, systems are characterized by their input-output relationships. See Figure 1 for a visual representation of signals and systems. A series of pop quiz questions below will help you reason about signals and systems.

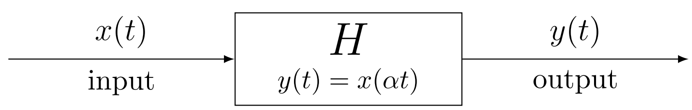


Figure 1: A system H that maps input signal $x(t)$ to output signal $y(t)$.

Pop Quiz 1.1: Check your understanding!

Based on the definitions above, if you were asked, "Is a matrix a signal or a system?" What would you say? Is the question well-posed? What would be the answer in context of the way matrices were introduced in the first lecture.

Solution on page 9

Pop Quiz 1.2: Check your understanding!

Can a system have multiple inputs? What about multiple outputs? Related, can a system change the domain of a signal?

Solution on page 9

Pop Quiz 1.3: Check your understanding!

When you apply filters on images on Instagram, identify the signals and the systems in that context (as many as you can!).

Solution on page 9

Pop Quiz 1.4: Check your understanding!

Can a function be a system? Can you pass in a function as an input to another function, and in that case, would the outer function be a system?

Solution on page 9

2 Key EE 102 idea: Break down complex signals into simpler signals

Since our goal in this course is to understand how signals can be broken down into simpler signals, we will need to understand the mathematical prerequisites that will help us do that. Let's start from the most basic equivalent concept — representing numbers as combinations of other numbers. In high school, you learned that any number can be represented as a linear combination of the numbers 1, 10, 100, etc. (the powers of 10). For example, the number 43 is actually $4 \times 10 + 3 \times 1$, that is, a linear combination of 10 and 1 with coefficients 4 and 3. This helps analyze numbers and perform arithmetic operations on them. Another approach you learned is prime factorization where you express any number as a multiplication of prime numbers. So, this general idea of representing something complex as a combination of simpler things is a very powerful and a very common mathematical idea. In this class, we need to do the same for signals. But before we get to things as complex as signals, let's take it one notch further from real numbers and let's talk about vectors.

2.1 Breaking down vectors as linear combinations of spanning vectors

We know that any vector in a given vector space can be represented as a linear combination of a set of spanning vectors. For example, in 2D space, any vector can be represented as a linear combination of the standard basis vectors $[1, 0]^\top$ and $[0, 1]^\top$. So, if you have a vector $[4, 3]^\top$, you can represent it as $4 \times [1, 0]^\top + 3 \times [0, 1]^\top$. This seems straightforward but it is very useful as now, you can just focus on analyzing the real coefficients 4 and 3, and the basis vectors are fixed.

We discussed how vectors can be used to represent quantities in a multi-dimensional space. Now we will discuss how complex numbers can be used to represent 2D vectors. This is particularly useful in signal processing, where we often deal with signals that have both magnitude and phase information (we will discuss what phase is in a few weeks). Overall, the central idea with complex numbers is that you want to carry two pieces of information in a single "number". This number is a complex number — it is complex because it is, after all, carrying more than one piece of information. More formally, we denote a notational convenience object "j" and say that the number can represent two quantities: one that is multiplied by 1 and the other that is multiplied by j. In this way, a complex number is storing two things but ensuring that they remain separate. Let's consider the example in vectors above, but now in complex number form.

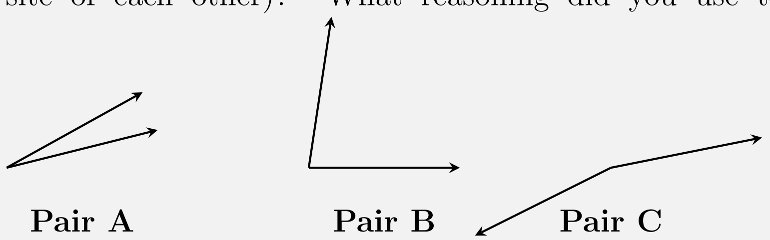
The vector $[4, 3]^\top$ can be represented as the complex number $4 + 3j$. The real part of the complex number (4) corresponds to the first component of the vector, and the imaginary part (3) corresponds to the second component.

2.2 Similarity between vectors and complex numbers

Whenever we talk about "breaking down something as a combination of other things" we are inherently defining how similar these objects are. So, mathematically, we need a clear notion of describing the similarity between the objects. For vectors, you have learned that the inner product (or the dot product) measures similarity between two vectors.

Pop Quiz 2.1: Check your understanding!

Out of the pairs of vectors shown below, which pair shows vectors that are most similar and which pair shows the vectors that are most dissimilar (opposite of each other)? What reasoning did you use to come to your conclusion?



Solution on page 9

2.3 Computing similarity: Inner products and linear dependence

In simple terms, the inner product between two vectors is a scalar quantity that quantifies a relationship between two vectors: how much they align with each other. In quantifying this, the inner product takes into account the lengths of the two vectors and the angle between them. The inner product can be used to define orthogonality (perpendicularity) — which is one the most fundamental concepts in signal processing.

Why? The key idea in EE 102 is that a linear combination of a set of orthogonal signals can be used to represent *any* signal (no matter how complicated), under some conditions, of course. So, understanding orthogonality, linear independence, and linear combinations is key to this course. Consequently, inner product is an important concept for this course.

If two vectors are orthogonal (that is, their inner product is zero), then these vectors are linearly independent. Indeed, we can prove that a set of non-zero mutually orthogonal vectors (say, $v_1, v_2, \ldots, v_n$) are linearly independent. You can show this by writing the linear combination of the vectors: $S = \alpha_1 v_1 + \alpha_2 v_2 + \ldots + \alpha_n v_n$ and showing that it is zero only if all constants α_i, $i = \{1, \ldots, n\}$ are equal to zero. To prove this linear independence, you can take the inner product of the above with any vector in the set (or any linear combination, thereof) say v_k:

$$S = v_k \cdot (c_1 v_1 + c_2 v_2 + ... + c_n v_n)$$
$$S = v_k \cdot c_1 v_1 + v_k \cdot c_2 v_2 + ... + v_k \cdot c_n v_n = 0$$

since the pairwise dot products (the inner product between each pair of vectors) are zero, we are only left with

$$c_k(v_k \cdot v_k) = 0$$

which is only possible if $c_k = 0$ since $v_k \cdot v_k$ is non-zero. Since this is true for any k, we have that all coefficients are zero. So, inner products play an important role in proving orthogonality (and thus, linear independence of vectors).

2.4 Inner product via complex numbers.

All the vector algebra can be tedious work! Complex numbers come to our rescue as we can represent a 2D vector as a complex number. For $\mathbf{x} = [x_1, x_2]^\top$ and $\mathbf{y} = [y_1, y_2]^\top$,

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = x_1 y_1 + x_2 y_2.$$

With the complex representations

$$z_{\mathbf{x}} = x_1 + \mathrm{j}x_2, \qquad z_{\mathbf{y}} = y_1 + \mathrm{j}y_2,$$

their product with conjugation is

$$\begin{aligned} \overline{z}_{\mathbf{x}}\, z_{\mathbf{y}} &= (x_1 - \mathrm{j}x_2)(y_1 + \mathrm{j}y_2) \\ &= (x_1 y_1 + x_2 y_2) \ + \ \mathrm{j}\,(x_1 y_2 - x_2 y_1). \end{aligned}$$

Taking the real part gives the vector inner product:

$$\mathrm{Re}(\overline{z}_{\mathbf{x}}\, z_{\mathbf{y}}) = x_1 y_1 + x_2 y_2 = \langle \mathbf{x}, \mathbf{y} \rangle.$$

Polar form view: Write $\mathbf{x}$ in polar coordinates with $r_x = \|\mathbf{x}\|$ and angle θ_x:

$$x_1 = r_x \cos\theta_x, \qquad x_2 = r_x \sin\theta_x,$$

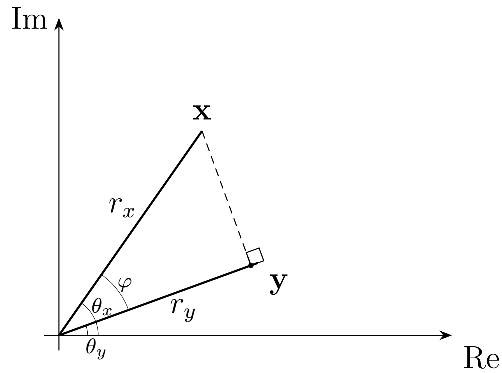


Figure 2: Geometric view of the inner product using polar form

so

$$z_{\mathbf{x}} = r_x(\cos\theta_x + \mathrm{j}\sin\theta_x) = r_x e^{\mathrm{j}\theta_x}.$$

Similarly $z_{\mathbf{y}} = r_y e^{\mathrm{j}\theta_y}$ from Euler's identity[1]. Then

$$\begin{aligned}\overline{z}_{\mathbf{x}}\, z_{\mathbf{y}} &= r_x e^{-\mathrm{j}\theta_x}\; r_y e^{\mathrm{j}\theta_y} = r_x r_y e^{\mathrm{j}(-\theta_x+\theta_y)} \\ &= r_x r_y \Big[\cos(-\theta_x+\theta_y) + \mathrm{j}\sin(-\theta_x+\theta_y)\Big],\end{aligned}$$

hence

$$\mathrm{Re}(\overline{z}_{\mathbf{x}}\, z_{\mathbf{y}}) = r_x r_y \cos(-\theta_x+\theta_y) = r_x r_y \cos(\theta_x-\theta_y) = \langle \mathbf{x}, \mathbf{y}\rangle.$$

Note that cosine is an even function, which allowed us to write the last equality above.

In Figure 2, observe that vectors $\mathbf{x}$ and $\mathbf{y}$ make angles θ_x and θ_y with the real axis and the angle between them is $\varphi = \theta_x - \theta_y$. You can make intuitive sense of the inner product in polar form by understanding its geometric interpretation (see Figure 2). Specifically, recall how we defined inner products in the previous section — a quantification of the alignment between two vectors. For our example in Figure 2, decompose $\mathbf{x}$ relative to $\mathbf{y}$: drop a perpendicular from the tip of $\mathbf{x}$ to the line $\mathrm{span}\{\mathbf{y}\}$ (the direction spanned by $\mathbf{y}$). This splits $\mathbf{x}$ into a part *parallel* to $\mathbf{y}$ and a part *perpendicular* to $\mathbf{y}$:

$$\mathbf{x} = \underbrace{(\mathbf{x}\cdot\hat{\mathbf{y}})\,\hat{\mathbf{y}}}_{\text{projection onto span}\{\mathbf{y}\}\text{, parallel to }\mathbf{y}} + \underbrace{\Big(\mathbf{x} - (\mathbf{x}\cdot\hat{\mathbf{y}})\,\hat{\mathbf{y}}\Big)}_{\text{perpendicular (rejection) to }\mathbf{y}}.$$

The second part that is perpendicular to $\mathbf{y}$ is called the rejection because that's the part that is remaining (you can see that it is quite literally the remaining part as it is obtained

[1] You can practice proving Euler's identity that $e^{\mathrm{j}\theta} = \cos\theta + \mathrm{j}\sin\theta$ by expanding the left-hand side using the exponential series and collecting the real and imaginary parts together (the real part will be the cosine series and the imaginary part will be the sine series).

by subtracting the projection from $\mathbf{x}$). Note that the inner product of $\mathbf{x}$ with the unit vector $\hat{\mathbf{y}}$ gives us the projection of x onto span$\{\mathbf{y}\}$. By computing the inner product, you can also check that the remaining (perpendicular part) of $\mathbf{x}$ is orthogonal to $\mathbf{y}$:

$$\operatorname{rej}_{\mathbf{y}}(\mathbf{x}) \cdot \hat{\mathbf{y}} = \mathbf{x}\cdot\hat{\mathbf{y}} - (\mathbf{x}\cdot\hat{\mathbf{y}})(\hat{\mathbf{y}}\cdot\hat{\mathbf{y}}) = \mathbf{x}\cdot\hat{\mathbf{y}} - \mathbf{x}\cdot\hat{\mathbf{y}} = 0.$$

Thus

$$\mathbf{x} = \operatorname{proj}_{\mathbf{y}}(\mathbf{x}) + \operatorname{rej}_{\mathbf{y}}(\mathbf{x}), \qquad \|\mathbf{x}\|^2 = \|\operatorname{proj}_{\mathbf{y}}(\mathbf{x})\|^2 + \|\operatorname{rej}_{\mathbf{y}}(\mathbf{x})\|^2.$$

In summary, the inner product measures how much of $\mathbf{x}$ points along $\mathbf{y}$, scaled by the length of $\mathbf{y}$. This is given by

$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{y}\| \, (\mathbf{x} \cdot \hat{\mathbf{y}}) = \|\mathbf{x}\| \, \|\mathbf{y}\| \cos\varphi.$$

The sign of $\cos\varphi$ carries the orientation: it is positive when the angle is acute and negative when obtuse. Complex numbers in their polar form express the same geometric intuition:

$$\overline{z_{\mathbf{x}}}\, z_{\mathbf{y}} = r_x r_y e^{\mathrm{j}(\theta_x - \theta_y)}$$

so the real part matches the dot product:

$$\operatorname{Re}\big(\overline{z_{\mathbf{x}}}\, z_{\mathbf{y}}\big) = r_x r_y \cos\varphi = \mathbf{x} \cdot \mathbf{y}.$$

Therefore, the scalar projection of $\mathbf{x}$ onto $\mathbf{y}$ is $r_x \cos(\theta_x - \theta_y)$, which when multiplied by the absolute value of y gives the inner product (that is, $r_x r_y \cos(\theta_x - \theta_y)$ in complex polar form).

2.5 Real-world significance

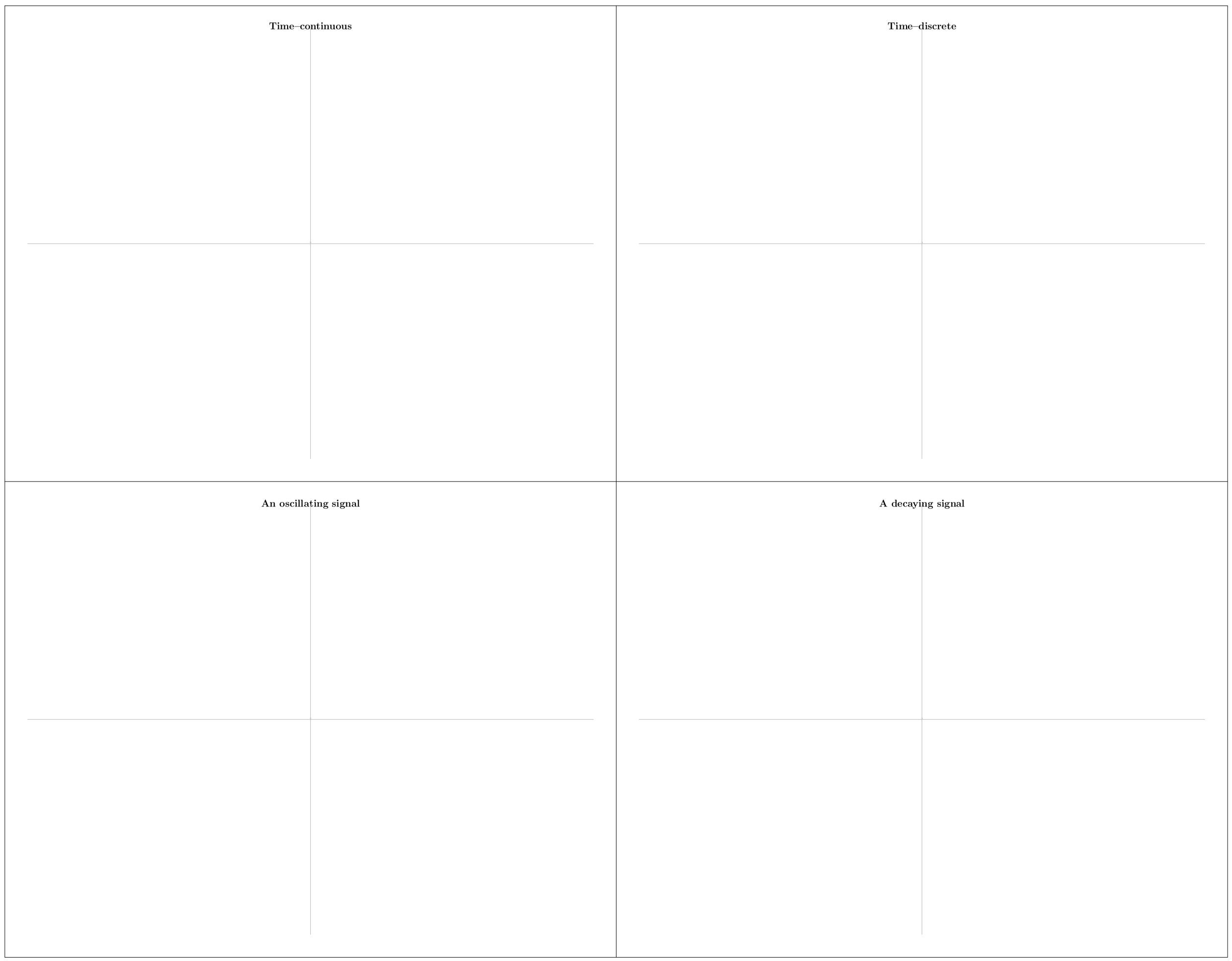
We discussed three main topics in this section — vectors, complex numbers, and their products. Representing quantities as vectors has many advantages, which mirrors the advantage of using lists and arrays in computer programming. Inner products are useful in quantifying the alignment between vectors. A simple example is in machine learning, where the similarity between data points can be measured using inner products, which has applications in face detection, recommendation systems, and more. Finally, as discussed, complex numbers help us analyze vectors in a more nuanced way by providing a framework for understanding their magnitude and direction. You will see many more real-world application examples of complex numbers in signal processing. In Fourier analysis, complex exponentials are used as the orthogonal basis functions for representing signals in the frequency domain.

3 Activity: Sketching signals

Before we study properties and types of signals, take some time to think about and sketch signals based on your intuition.

Worksheet #1: Sketching Signals (Part 1) — Groups of 2.

Each student takes one quadrant. Label axes clearly and annotate *what is your signal?*, *where is the signal likely to be observed?*, and key *properties of the signal*.



Pop Quiz Solutions

Pop Quiz 1.1: Solution(s)

Matrices as transformation operators are systems, they transform a vector into another vector. However, matrices can also be used to represent signals, for example, a matrix of pixel values in an image signal.

Pop Quiz 1.2: Solution(s)

Yes, a system can have multiple inputs and multiple outputs. A system can also change the domain of a signal, for example, by converting a continuous-time signal to a discrete-time signal.

Pop Quiz 1.3: Solution(s)

The input image is a signal, the filter that you are applying is a system because it maps the input image to an output image. The output image is also a signal. Internally, the filtering app might be passing the image through other systems, like passing it through to the native image displaying app in the correct resolution to display it back to you.

Pop Quiz 1.4: Solution(s)

It depends on what you mean by mathematical function. Usually, we say that mathematical functions that represent a physical quantity would be a signal. A system, on the other hand, is a process that transforms one signal into another.

Pop Quiz 2.1: Solution(s)

Pair A shows the most similar vectors because the angle between the two vectors is the smallest. Pair C shows the most dissimilar vectors because the vectors point in nearly opposite directions, so the angle between them is the largest.
One way to quantify this is using the dot product:

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \, \|\mathbf{b}\| \cos\theta.$$

After accounting for the lengths of the vectors, vectors pointing in similar directions have $\cos\theta$ close to 1, while vectors pointing in opposite directions have $\cos\theta$ close to -1.